UNIT - 4.

* Schema Refinement

It is the process of refining the schema so as to solve the problems caused by redundantly storing information, (duplication of data) this can be done by decomposing larger tables into smaller ones.

* **Redundancy** :- the duplicates present in the code or the repeated data is called redundancy.

- **Problems caused by Redundancy** :-

1. Leads to wastage of memory space.
2. Possibility of inconsistent data.
3. It leads to more difficult data updates.
4. Certain anomalies.

* **Anomalies** :- these are the problems that occur in unnormalized databases.

The following types are existing :

1. Insertion Anamoly.
2. Deletion Anamoly
3. Updation Anamoly.

* **Insertion Anamoly** :- let us assume that a new department was started by a particular organisation and initially there is no employee in that department then the tuple for that department cannot be inserted into the table because of some other constraints. This kind of certain ~~records~~ tuple cant be inserted into table is called insertion Anamoly.

Deletion Anamoly :- consider a situation where only 1 employee is working in the dept and that employee leaves the organisation then we have to remove his details from the database but in addition to his information the information about the dept is also lost. This is deletion anomaly: whereas certain information is lost because of deletion of other information

Updation Anomaly:-

* Normalization :-

- It is a step-by-step process to which we can decompose or divide a bigger relation into more than one relation

Types of Normal Forms :-

1. First Normal Form (1NF)
2. Second Normal Form (2NF)
3. Third Normal Form (3NF)
4. Boycce codd Normal Form (BCNF or 3.5 NF)
5. Fourth Normal Form (4NF)
6. Fifth Normal Form (5NF)

Functional Dependency :- It is a relationship where the value of one/ more attributes uniquely determines the value of another attribute. It is represented as $x \rightarrow y$ where x is the determinant and y is dependent

Consider a student with roll no, name, age, deptname as it's attributes. Write the 2 Functional Dependencies from above schema.

stud (rollno, name, age, deptname)
1 x 20
2 x 21

rollno → name ✓

rollno → age ✓

$\boxed{\text{rollno} \rightarrow \text{deptname}}$ × did can be used to identify dname uniquely.

Types of Functional Dependencies :-

1. Partial Depedency.
2. Fully Functional Dep
3. Trivial Functional Dep
4. Non-Trivial Functional Dep.
5. Transitive Functional Dep.
6. Multi-Valued Functional Dep

→ Partial Dependency : Consider an FD $XY \rightarrow Z$ and assume that Z can be determined using any one of the determinants that is either X or Y. This leads to the concept of Partial functional dependency. $XY \rightarrow Z$

→ Fully Functional Dependency: Consider an FD $XY \rightarrow Z$ and assume that Z can be determined by using both the determinants (XY). This is the concept of Fully Func. Dep.

$\boxed{XY} \rightarrow Z$

→ Trivial Functional Dependency:- Consider an FD $X \rightarrow Y$ if Y is a subset of X then it is called a trivial func dep.

$X \rightarrow Y$ $Y \subseteq X$ $\boxed{XY \rightarrow X}$ → X is a subset of XY.

XY should determine X

Consider a table

Users info (uid, uname, email) write following procedures

1) Insert a new user using procedures
2) Display information about all the users.
3) Count the total no. of users existing in the user info table
4) Update email id of a particular user given uid.
5) Write a procedure to find whether a no. positive, negative or equal to 0.

- create table users(uid int primary key, uname varchar(20), email varchar(20));

- insert into users values (

SVD

cid	cname
01 →	x
02	y
03	z
01	x ✓
$\boxed{01 \to y}$	✗

Each determinant got one dependent

MVD min = 3 cols.

sid	Hobbies	sports
01	H_1	S_1
01	H_2	S_2
02	H_2	S_3
02	H_3	S_5
03	H_3	S_4

Violation of 3NF — TD — 3 cols

$\boxed{A \to B,\ B \to C,\ A \to C}$

* **SVD** :- Single valued Dependency.

→ It is a constraint where for a specific value or set of values of one attribute there is precisely one corresponding value for another attribute. cid implies cname

FD → $\boxed{\text{cid} \longrightarrow \text{cname}}$

* **MVD** :- Multi Valued Dependency:

→ MVD occurs in a database when one attribute in a table determines a set of values of another table & this set of values is independent of another ~~value~~ attribute in the table then MVD exist. Represented by $X \twoheadrightarrow Y$

* **Transitive Functional Dependency (TD)**:

→ Here, a dependent is indirectly dependent on the determinant. i.e $A \to B$, $B \to C$ then we have $A \to C$

→ Minimum no of columns for TD exist is 3.

~~sid~~ ~~sname~~ marks ~~sid~~ ~~marks~~

~~1~~ ~~A~~
~~2~~ ~~B~~
~~3~~ ~~C~~

sid	dept	loc	add
1	IT	B_1	Hyd
2	CSE	B_2	Hyd
3	CSE	B_2	Hyd
4	IT	B_1	Hyd
5	DS	B_3	Hyd

sid → dept
dept → loc
sid → loc

non prime key is determining other non prime key attr. → TD

* Armstrong's Axioms :-

→ It refers to a set of inference rules & it was given the name as it was introduced by William W. Armstrong

Basically divided into 2 cat

1. Primary Rules	2. Secondary Rules
• Reflexivity	• Union
• Augumentation	• Decomposition
• Transitive	• Pseudo Transitivity
	• Composition Rule.

Primary Rules.

- **Reflexivity** : If a is set of attributes and b is a subset of a then a holds b. [$a \to b$] $b \subseteq a$
- **Augumentation** : If $a \to b$ holds good and x is a set of attributes then $ax \to bx$ also holds good. (i.e adding attributes to an existing dependency doesnot change existing dependency)
- **Transitive** : same as transitive rule in basic algebra i.e $a \to b$ holds & $b \to c$ holds then $a \to c$ also holds good.

Secondary Rules

- **Union** :- $A \to B$ $A \to C$ then $A \to BC$
- **Decomposition** :- $A \to BC$ then $A \to B$, $A \to C$

  $\boxed{AB \to C \quad \begin{matrix} A \to C \\ B \to C \end{matrix} \times}$ → wrong.
- **Composition** :- If $A \to B$ and $X \to Y$ hold then we can say $AX \to BY$ holds.
- **Pseudo Transitivity** : If $A \to B$ and $BC \to D$ then $AC \to D$.

* Attribute Closure (AC) :- for a set of attributes X is the set of all the attributes that can be functionally determined from X using a given set of functional dependencies. It is represented as X^+.

Uses of AC :-

1. Used in determining the candidate keys etc.
2. Checks if the F.D holds.
3. Check lossless Decomposition
4. Check dependency preservation

R(ABCDE)

FD = {A→B, B→C, AC→D, D→E}

A^+ = ABCDE

B^+ = BC

BD^+ = BDCE

AC^+ = ACDEB

Consider a Relation R(A B C D E F)

FD = {E→A, E→D, A→C, A→D, AE→F, AF→E}.

Find E^+ = EADCF

~~Uses of C No~~

Finding keys given a Relation R & set of FDs

R(ABC)

F = {A→B, B→C}

A^+ = ABC [S.K]

B^+ = BC X

C^+ = C X

AB^+ = ABC [S.K] ABC C

A^+ ✓ B^+ φ

ABC BC

BC^+ = BC

AC^+ = ACB

A^+ C^+ φ

here super keys → [AB, A, AC, ABC]

candidate key → A [minimal subset of S.K]

Consider R(ABCDEF)

F: {AB→C, C→DE, E→F, D→A, C→B}

$(ABCDEF)^+$ = ABCDEF (S.K)

using 1 AB→C

$(ABDEF)^+$ = AB DEF C (S.K)

using 2 C→DE

$(ABF)^+$ = ABCDEF (SK)

AB^+ = ABCDEF (SK)

A^+ B^+

A B → prime attributes

AB is a C.D [minimal of SK]

Incomplete

Uses of Normalization

1) eliminating redundant data
2) helps in keeping data consistent
3) Storage optimization
4) Breaking large tables into smaller tables with relationships
5) Making database more scalable & adaptable.

1NF :- For a relation to be in 1NF it should hold - props:

- Table should have single valued attributes
- Values stored in a column should be of same domain (int means int only.)
- All the columns in a table should have unique names (sid, sage, sname ≠ sage)
- The order in which data is stored doesnot matter. (01, 05, 02 ✓)

Ex

sid	sname	sage	Mob
1	A	20	M1, M2, M3
2	B	21	M1, M2
3	C	22	M1

(x) Multivalued

Method 1

Foreign key → can be repeated

R_1 (sid, Mob)

sid	Mob
01	M1
01	M2
01	M3
02	M1
02	M2
03	M1

R_{11} (sid, sname, sage)

sid	sname	sage
1	A	20
2	B	21
3	C	22

✓ 1NF

Method 2

sid	sname	sage	Mob1	Mob2	Mob3
1	A	20	M1	M2	M3
2	B	21	M1	M2	Null
3	C	22	M1	Null	Null

2NF :- a relation R is said to be in 2NF if it satisfies the following conditions:

1) R should be in 1NF
2) There is no concept of Partial Dependency.

Relation & set of FDs

$A \rightarrow B$
$AC \rightarrow B$

① Candidate key
② Prime Attributes, Non-Prime Attributes
③ look for any PD (FDs)
  - Yes - Not in 2NF
  - No - In 2NF.

Find R is → 2NF

R(A,B,C,D)

$AB \rightarrow CD$, $C \rightarrow A$, $D \rightarrow B$.

Find R is in 2NF

R(A,B,C,D)

$AB \rightarrow CD$, $C \rightarrow A$, $D \rightarrow B$

$AB^+ \rightarrow ABCD$

$C^+ \rightarrow AC$

$D^+ \rightarrow BD$

$ABC \rightarrow C \cdot D$

$AB^+ \to A, B, C, D$ S.K

$A^+ \Rightarrow A, D, C$

$B^+ \Rightarrow B, D, C$

A → C, B → D

$(ABCD)^+ = ABCD$ S.K

$(AB)^+ = ABCD$ S.K

other relations cannot decompose so,

$(AB)^+$

empty set ← ϕ A^+ B^+

A   B

$C \to A$: AB, CB

$D \to B$: AB, AD

C.D ⇒ AB, CB, AD, CD.

P.A ⇒ A, B, C, D

Given a relation $R(A,B,C,D,E,F)$ $[A \to B, B \to C, C \to D, D \to E]$

$(ABCDEF)^+ = ABCDEF$ S.K $\Longrightarrow$ $(A\cancel{B}\cancel{C}\cancel{D}\cancel{E}F)^+$

$(AF)^+$

$A^+ = AB$ $ACDEF = ABCDEF$ S.K

$B^+ = BC$ $ADEF = ABCDEF$ S.K

$C^+ = CD$ $AEF = ABCDEF$

$D^+ = DE$ $\boxed{(AF)^+ = ABCDEF \text{ S.K}}$

ϕ A   F

× ABCDE   F ×

So AF is C.D key

P.A ⇒ A, F   N.P.A ⇒ B, C, D, E

$A \to B$

B (npa) is depedent on a part of C.D (A) so violation of 2NF cond.

Consider $R(A,B,C,D)$ $[A \to B, B \to C, C \to D]$

$(A\cancel{B}\cancel{D})^+ = ABCD$

$A^+ = ABCD$

$A \to$ C.D, P.A ⇒ A, N.P.A = B, C, D

∴ In 2NF ✓

$R(A,B,C,D)$ $[A \to B, B \to C]$. Find R is in 2NF

$(A\cancel{B}\cancel{C}D)^+ \Rightarrow ABCD$.

$(AD)^+ \Rightarrow ABCD$.

A^+ D^+

ABC ×   D ×

$AD^+ \Rightarrow$ C.D key.

P.A ⇒ A, D

N.P.A ⇒ B, C

$A \to B$ ⊗ ⟶ Not in 2NF

$B \to C$ ✓

Consider $R(A,B,C,D)$ if not in 2NF decompose.

$[AB \to D, B \to C]$

↓

Not violating

$(AB\cancel{C}\cancel{D})^+ \Rightarrow ABCD$

ϕ $AB^+ \Rightarrow ABCD$

A^+: A, D B^+: BCD

$B \to C$ P.D

∴ Not in 2NF

$R_1(A, \underline{B}, D)$

$R_2(\underline{B}, C)$

Did	Pid	Pname	Price	qty
1	101	Mouse	500	2
1	102	Keyboard	800	1
2	101	Mouse	500	4

AB^+ — ABCDE

$AB \to CD$ ABCDE

$D \to E$

Did, Pid

C.D → (Did + Pid)

decompose: the table if not in 2NF.

* **3NF :-** a relation R is said to be in 3NF if the following conditions hold.

1) R should be already in 2NF
2) No non-key attribute is transitively dependently on key attribute

(or) No non-prime key attribute functionally determines another non-prime key attribute.

Consider a relation R (A, B, C, D) $A \to B$, $B \to C$, $C \to D$ as FD's. Find whether R is in 3NF. (Assume R in 1NF)

$ABCD^+ = ABCD$.   P.A = A (simple attribute)

$A^+ = ABCD$ (S.K)   N.P.A = B, C, D

(C.D).

∴ In 2NF

3NF → $A \to B$ ✓ $B \to C$ $C \to D$. ⟹ NOT IN 3NF.

N.P.A → N.P.A ✗

R (A, B, C, D, E, F) $AB \to CDEF$, $BD \to F$

P.A N.P.A → NPA ✓ valid

$ABCDEF^+ = ABCDEF$ C.D = AB

$AB^+ = ABCDEF$.   P.A = A, B

A^+ ϕ B^+ N.P.A = C, D, E, F

ACDEF   BDF

	2NF	3NF
$AB \to CDEF$	✓	✓
$BD \to F$	✓	✗

In 2NF But not in 3NF

$R(A, B, C, D, E, F)$

$AB \to CDEF$, $BD \to F$, $A \to D$.

$ABCDEF^+ = ABCDEF$

$AB^+ = ABCDEF$

$A \quad B$

$DB = ABCDEF$.

C.D = AB, ~~DB~~.

P.A = A, B, ~~D~~

N.P.A = C, E, F, D.

	2NF	3NF.
$AB \to CDEF$	✓	✗
$BD \to F$	✓	✗
$A \to D$	✗	✗

not in 2NF so not in 3NF.

sid	sname	DoB	state	country	pin	total_credits
1	A	-	TS	IN	1	-
2	B	-	TS	IN	1	-
3	C	-	TS	IN	1	-
4	D	-	AP	IN	2	-

candidate key - sid

1NF - Yes [no multivalued attributes]

2NF - Yes [no partial dependency] only one c.k.

3NF - No [pin (npA) → country / state (npA)].

$R(A, B, C, D, E)$ $A \to B$ $B \to C$ $C \to D$, $D \to A$

$ABCDE^+ \to ABCDE$

$AE^+ \to ABCDE$

$A^+ \quad E^+$

ABCD $\quad$ E

AE

DE

C.DK ⇒ AE, DE

P.A ⇒ A, D, E

NPA ⇒ B, C

	2NF
$A \to B$	✗
$B \to C$	✓
$C \to D$	
$D \to A$	

Not in 2NF $\boxed{\therefore \text{ not in 3NF.}}$